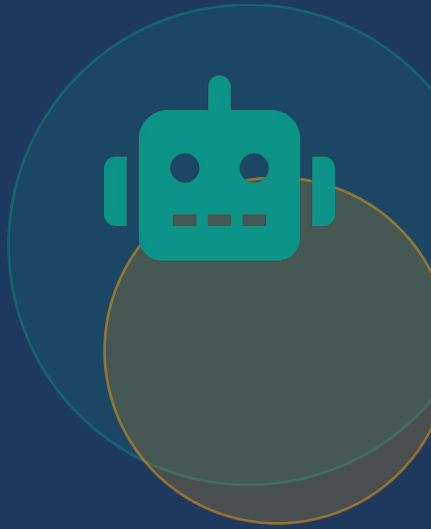
MIDTERM ASSIGNMENT

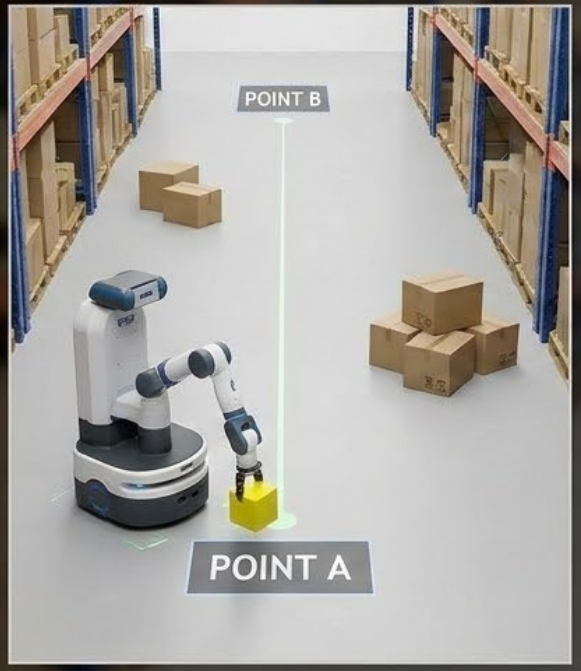
Digital Twin & Robot Simulation

From Scene Reconstruction to AMR Task Execution



NVIDIA Isaac Sim · 3D Gaussian Splatting · AMR Navigation · Pick & Place

ISAAC SIM SIMULATION: AMR PICK & PLACE WITH DOMAIN RANDOMIZATION



SCENARIO 1:
BASELINE

SCENARIO 2:
RANDOMIZED OBSTACLES

SCENARIO 3:
GOAL & ENVIRONMENT VARIATION

Basic

30%

Navigation & Obstacle Avoidance

- AMR reaches the target point
- Successfully avoids all obstacles
- Start & end points randomized per run

30%

Arm Operation (Pick & Place)

- Successfully grasps the object
- Successfully places the object
- No precision requirement at this stage

20%

Domain Randomization

- Randomize AMR start position
- Randomize AMR goal position
- Randomize object & placement location

20%

Technical Report & Video

- System architecture overview
- Coordinate setup documentation
- Video of ≥3 randomized runs

Bonus Items

+3%

3DGS Scene Reconstruction

- Capture real-world scene with camera
- Complete 3DGS reconstruction
- Import into Isaac Sim & run simulation
- Reconstruction quality is NOT evaluated

+ 2%

Grasp 3DGS-Reconstructed Object

- Reconstruct a real object using 3DGS
- Replace standard CAD model with it
- Set up collision proxy (convex hull)
- Successfully grasp & place the object

+5%

Robot Diversity

- Replace standard robot chassis
- Use custom base (e.g., quadruped)
- Or use custom end-effector / gripper

Submission Deadline: 5/11

Required

Project Report (PDF / Word)

Isaac Sim setup screenshots, Domain Randomization implementation notes, troubleshooting log

Demo Video

At least 3 randomized execution runs, with Physics Collider visualization enabled

Code & Assets

Python scripts (path planning, control, randomization logic) and scene USD file

Optional (Bonus)

3DGS Output (Optional)

Reconstructed scene & objects, import screenshots, collision proxy setup explanation

Evaluation uses official USD scene + Nova Carter + Franka Panda. Standard cube object provided for consistent physics parameters.